



Effect of **hydrological events** on the **steady/unsteady** swimming in Salmonidae (*Salmo trutta*)

GUILLAUD Felix



“Global change causes augmentation in **floods** and **droughts**’ intensity and frequency in rivers”

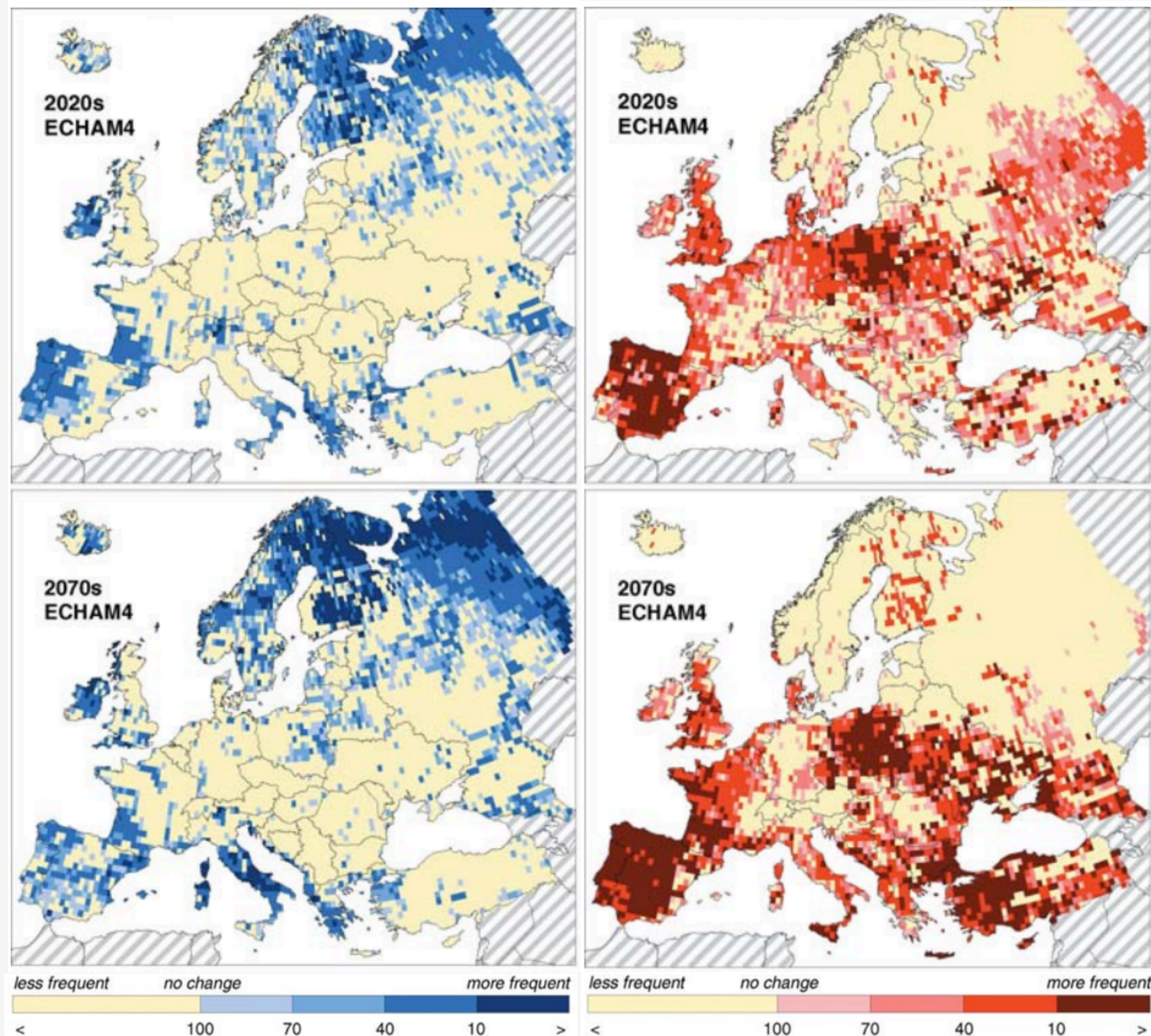


Figure 1: Change in recurrence of 100-year floods (blue) and droughts (red), based on comparisons between today's climate and water use (1961–1990) and simulations for the 2020s and 2070s (ECHAM4 climate models). From Lehner *et al.*, 2006



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How does **hydrological events** impact the **morphology** and the **performances** of river fishes ?

Water velocity shapes juvenile salmonids

Behaviour and growth of juvenile brown trout exposed to fluctuating flow

LUNAR LUNDS

Predictability of phenotypic differentiation across flow regimes in fishes

R. Brian La
*Museum of
Cambridge, M

Effect of the Flow Turbulence on the Movement Pattern of the Caudal Fin in Fish

The effect of turbulence on the cost of swimming for juvenile Atlantic salmon (*Salmo salar*)¹

Eva C. Enders, Daniel Boisclair, and André G. Roy

Kinematics / Morphology.

Swimming trade-off

manoeuvring
vs.
stability

Ondulatory swimming

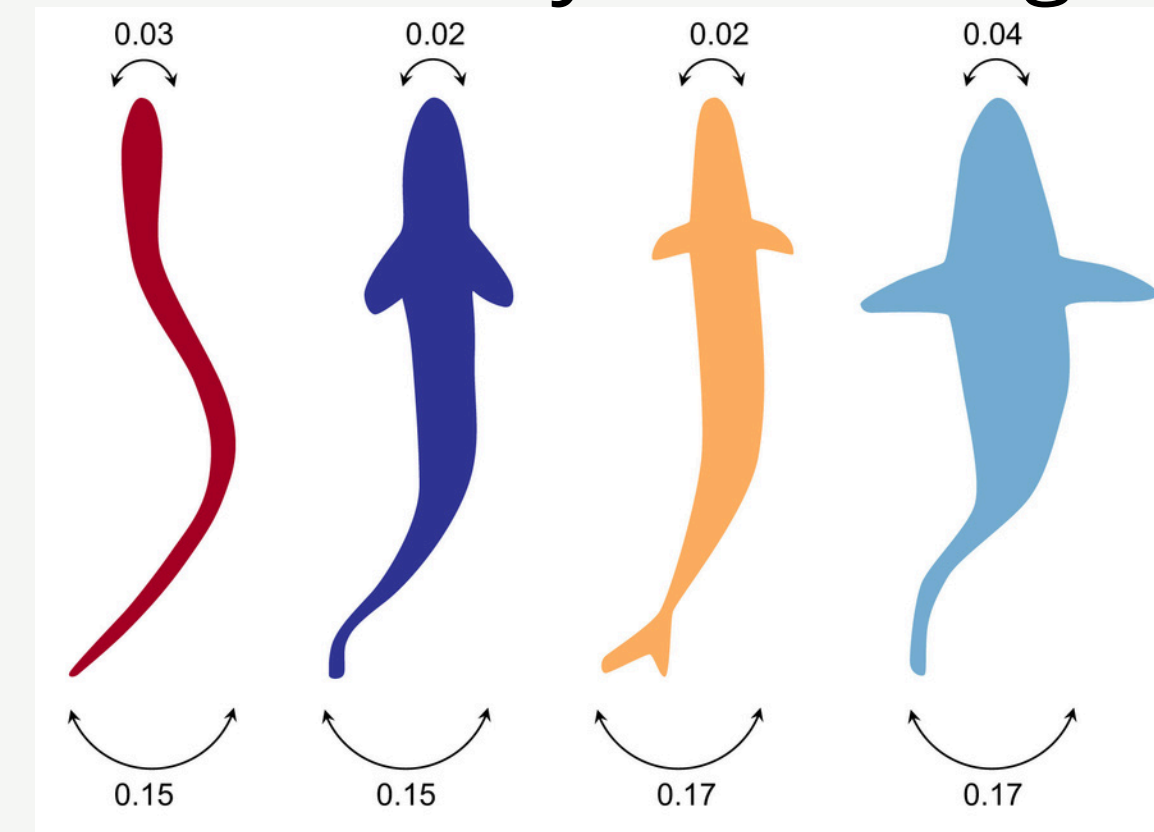


Fig. 2. Illustration of the four classic fish swimming modes based on high-speed video analysis of steady swimming kinematics. eel, trout, mackerel and tuna.. From Di Santo *et al.*, 2021

→ macro-evolutive scale

Kinematics / Morphology at a micro-evolutive scale

Ondulatory swimming

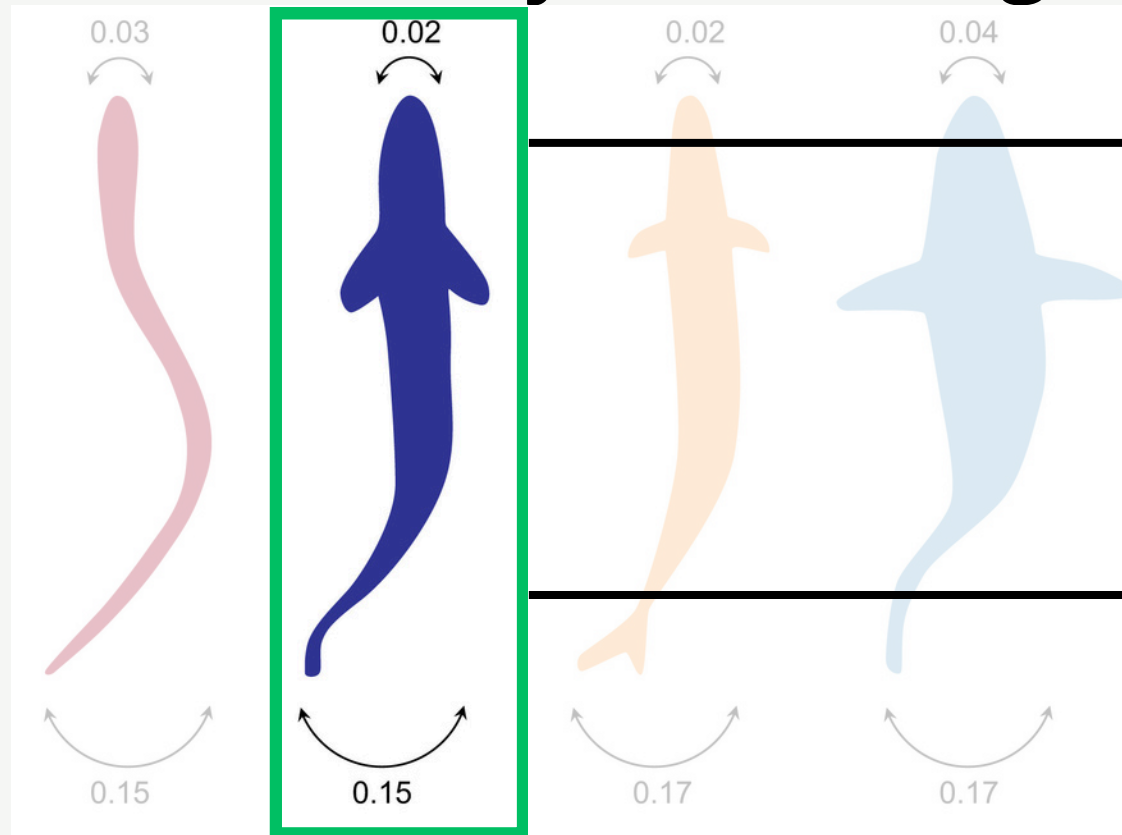


Fig. 2. Illustration of the four classic fish swimming modes based on high-speed video analysis of steady swimming kinematics. eel, trout, mackerel and tuna.. From Di Santo *et al.*, 2021

→ morphology (ongoing PhD project)

→ **swimming performances**

→ macro-evolutive scale

→ micro-evolutive scale



Objectives:

Impact of **hydrological events** (floods) on swimming **performances** of *Salmonidae*

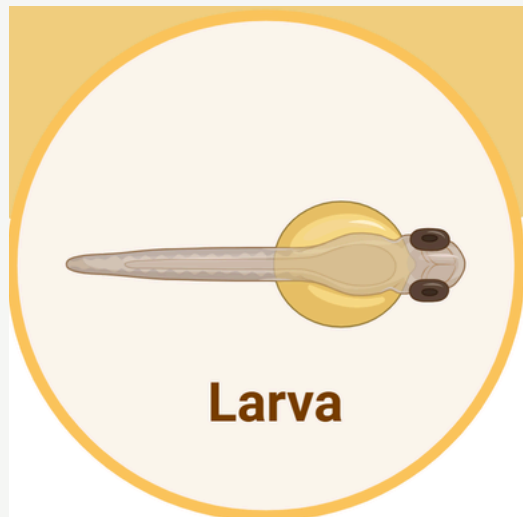
- ➔ How does it impact fish' swim and capacity to move through different flow regimes ?
- ➔ Can we detect typical differences in swimming / behavioural syndrome ?
- ➔ Can we expect typical differences in swimming trade-off ? (stability/manoeuvrability)

Biological model: *Salmo trutta*

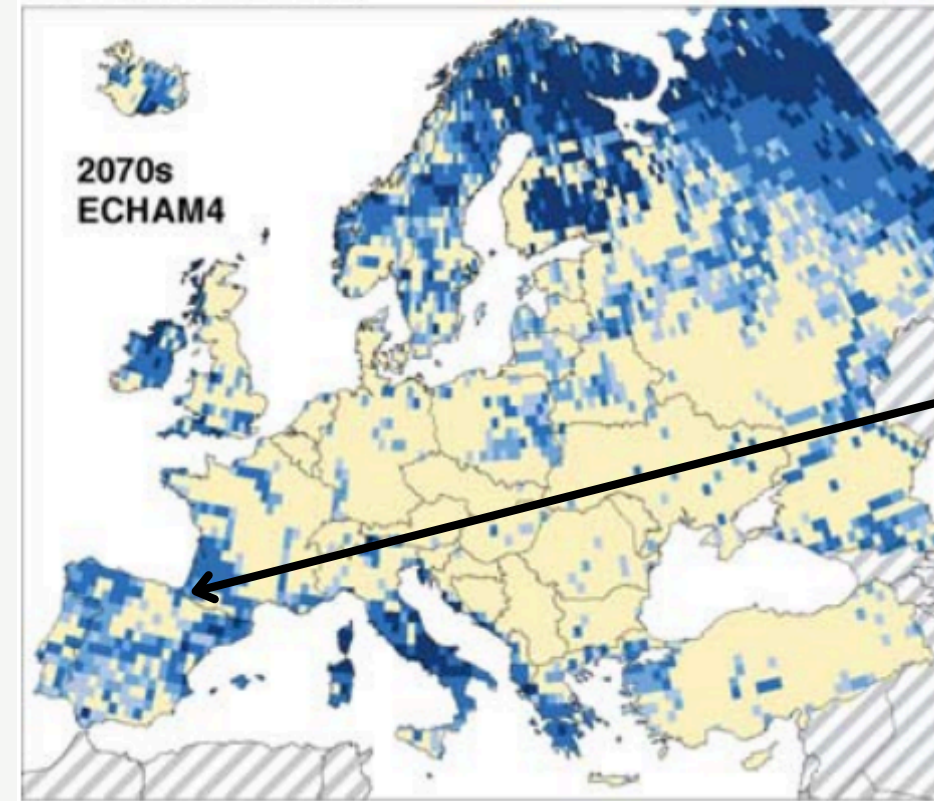
Common biological model
Migration Aquaculture **Kinematics**



- most of the morphological variability during ontogeny
- micro-evolutive scale
- abundant in the region



Site:



aerial view of the experimental river © MARQUOT

Treatment:

Upstream treatment: Varying flow, intensity up to 5x the normal flow, 2 to 3 times a week

Downstream treatment:
Low variation flow (~constant)

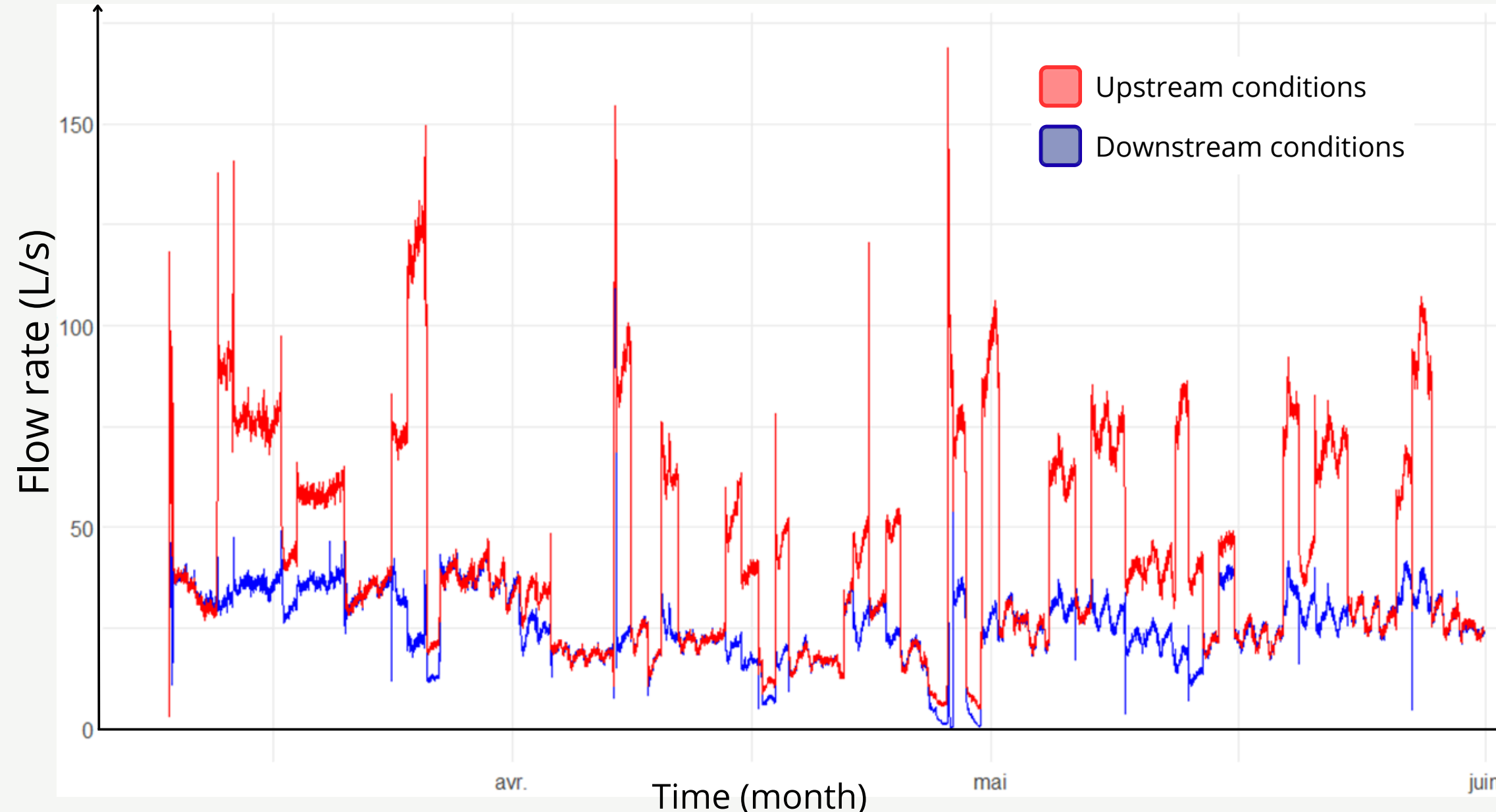
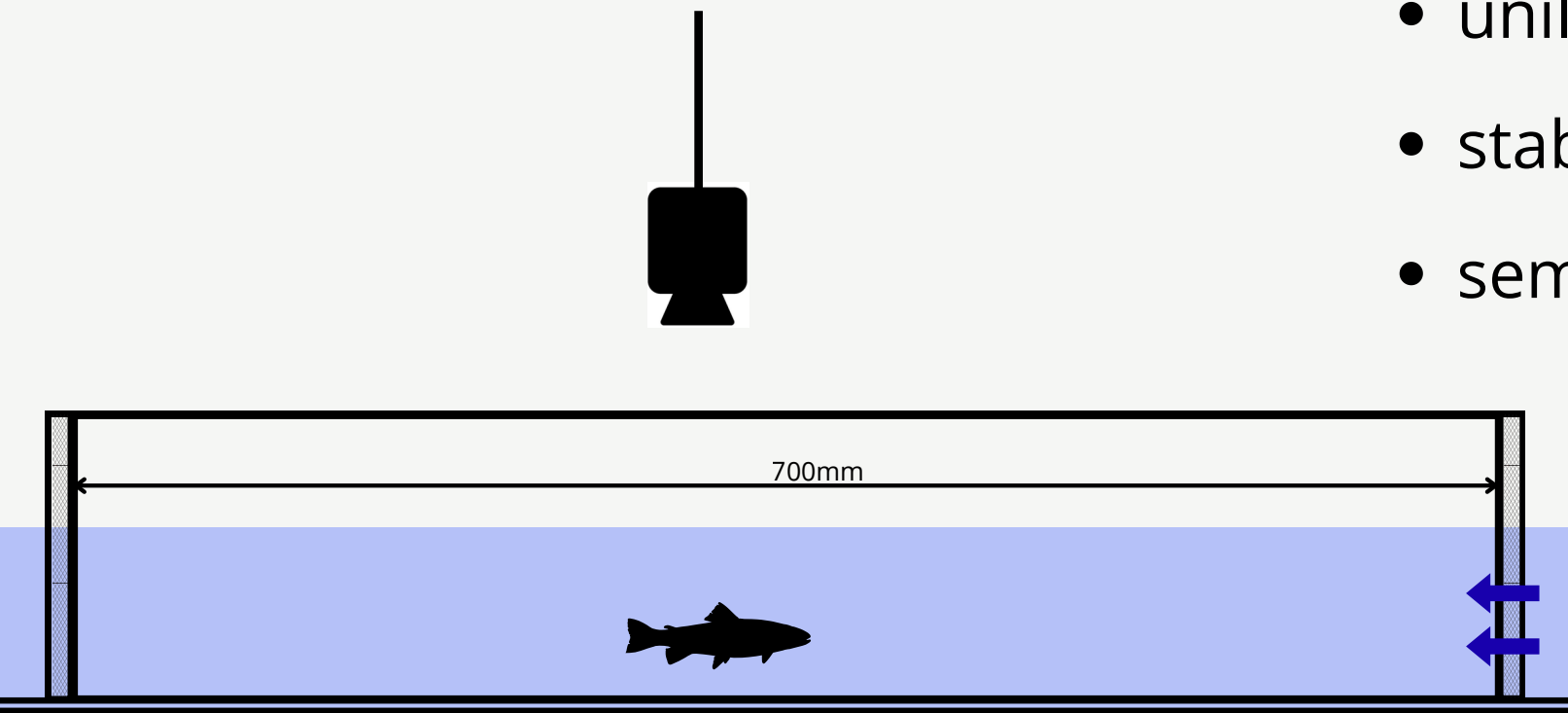


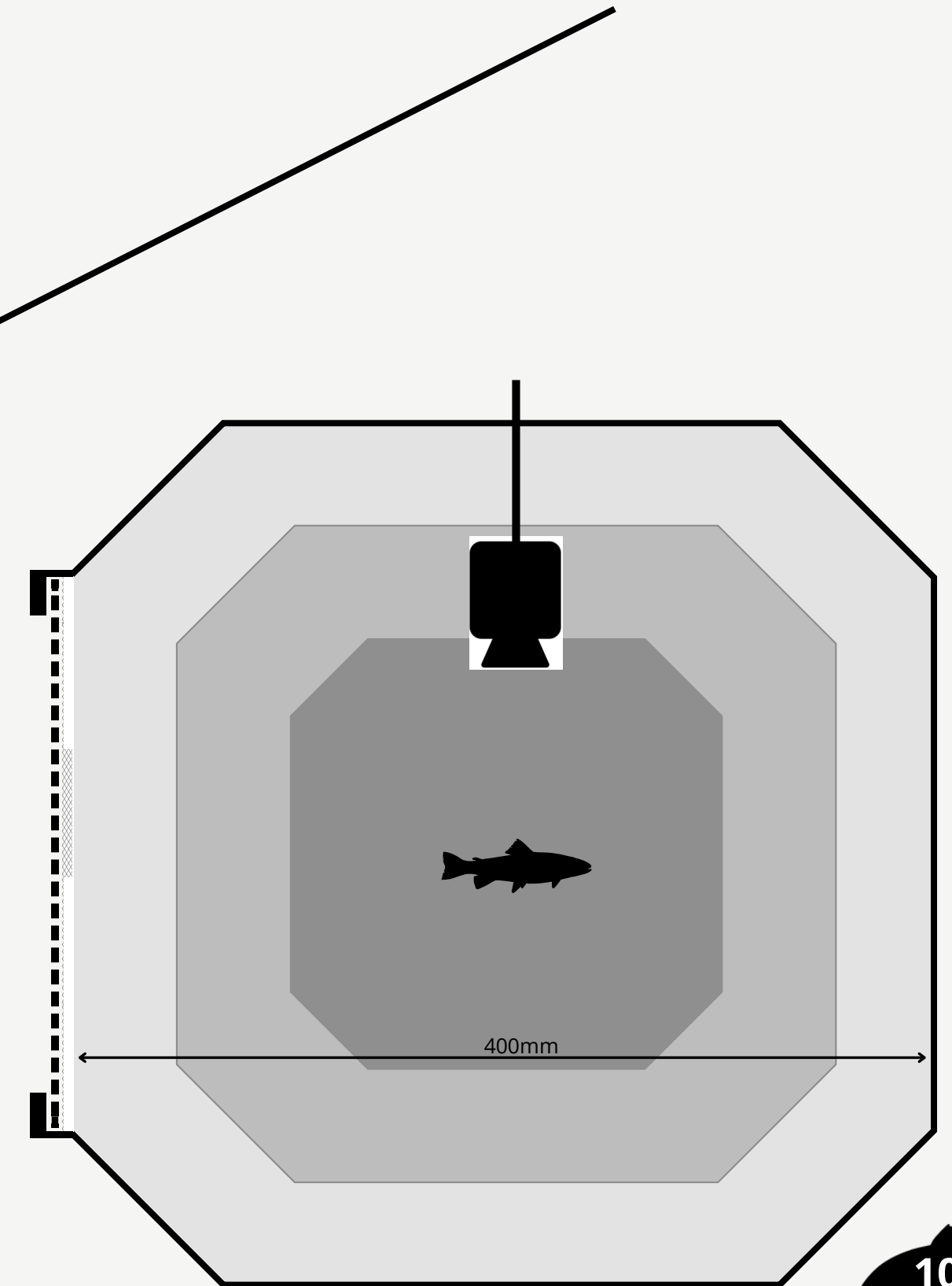
Figure 3: Illustration of the variation of flow rate during the 3 months of exposure to treatment inside the artificial river with fluctuant conditions (upstream) in red and stable conditions in blue (downstream).

Swimming tests:**Steady swimming analysis**

- unilateral constant flow
- stability performance
- semi-natural conditions

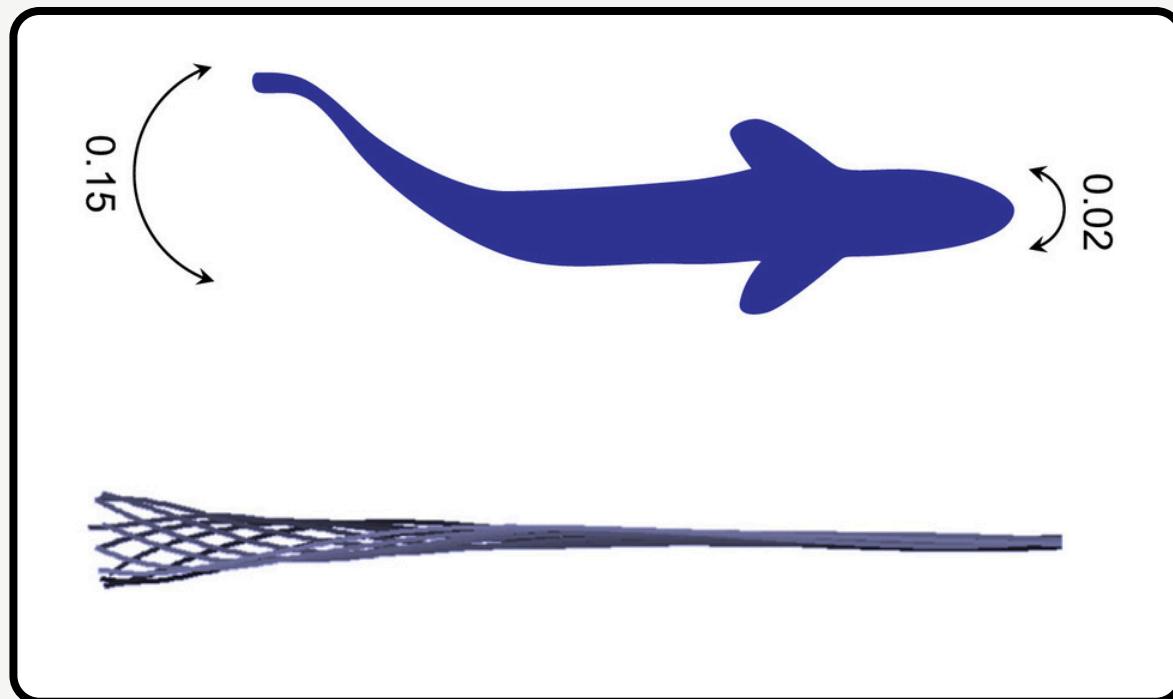
**Unsteady swimming analysis**

- no flow regime
- behaviour and escape responses
- manoeuvring performance
- artificial conditions



Measures:**Steady swimming**

e.g.

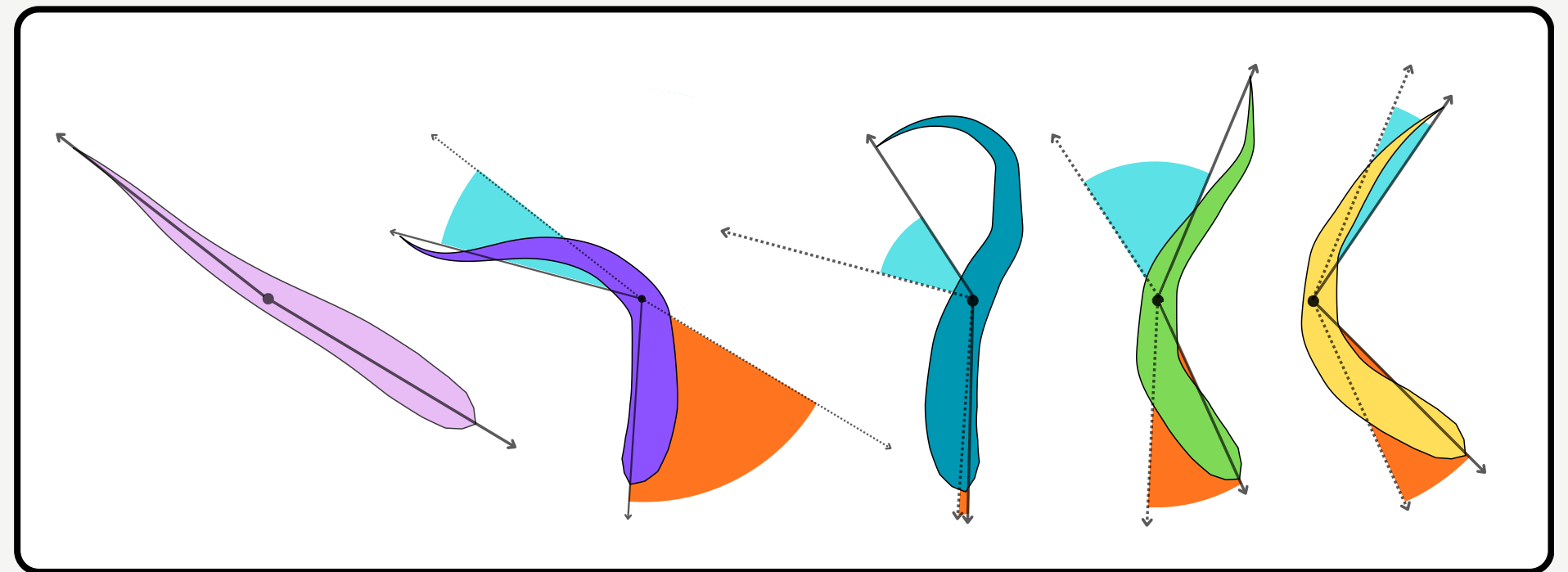


(tail-beat frequency, amplitude)
stability/cruising performance

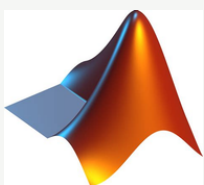
Unsteady swimming

Behaviour + Fast-start

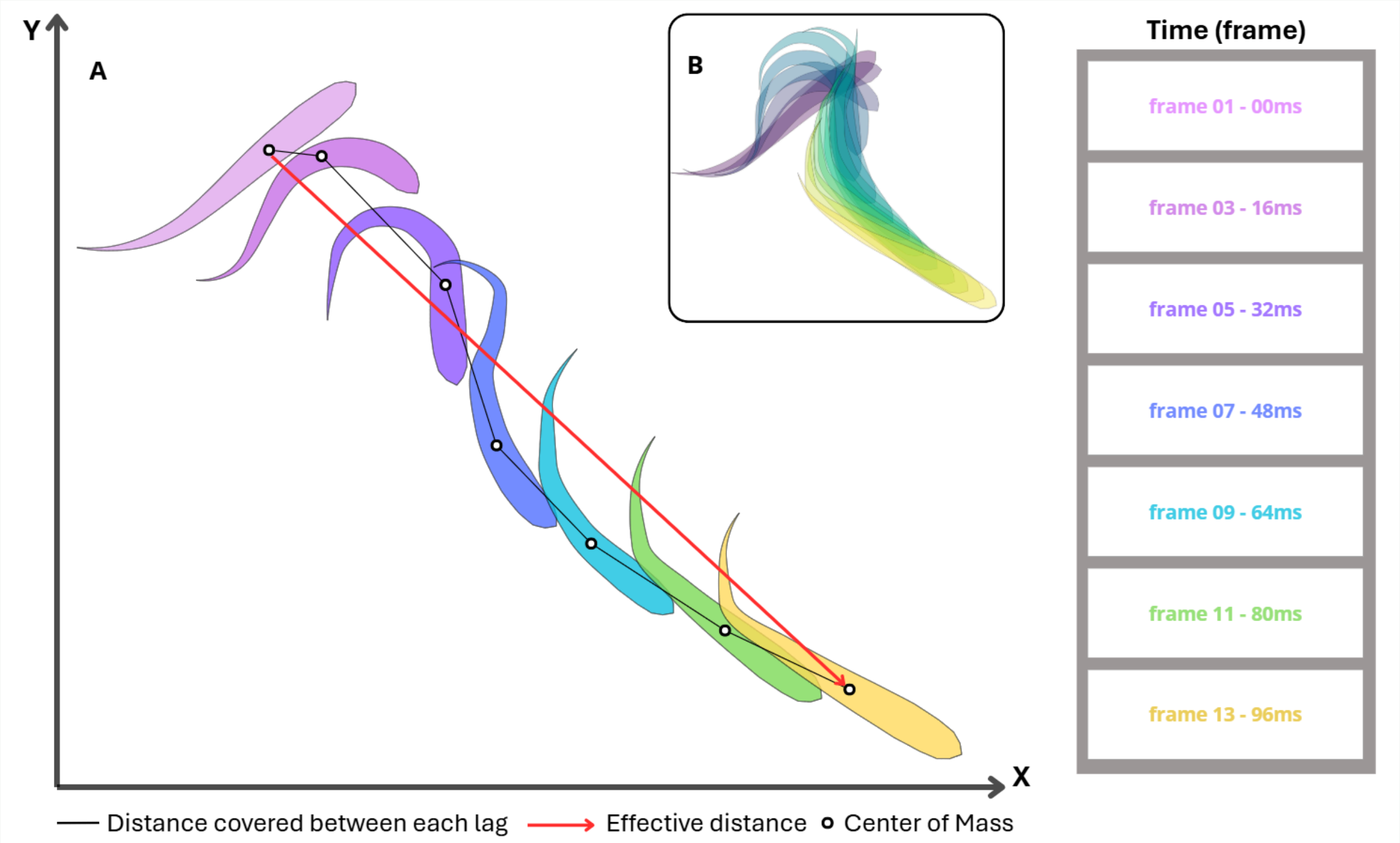
e.g.



(distance, acceleration, efficiency, activity...)
manoeuvring performance



e.g. Efficiency during fast-start



efficiency = $\frac{\text{sum of distance covered}}{\text{effective distance}}$

Figure 4: Illustration of the distance covered by the fish during a fast-start event with 16ms lag and the representation of the effective distance (red arrow). **A)** Fish actuation through space with colour graduation for time. **B)** Silhouettes of the fish during the 14 frames of the fast start



Body midline digitalization:

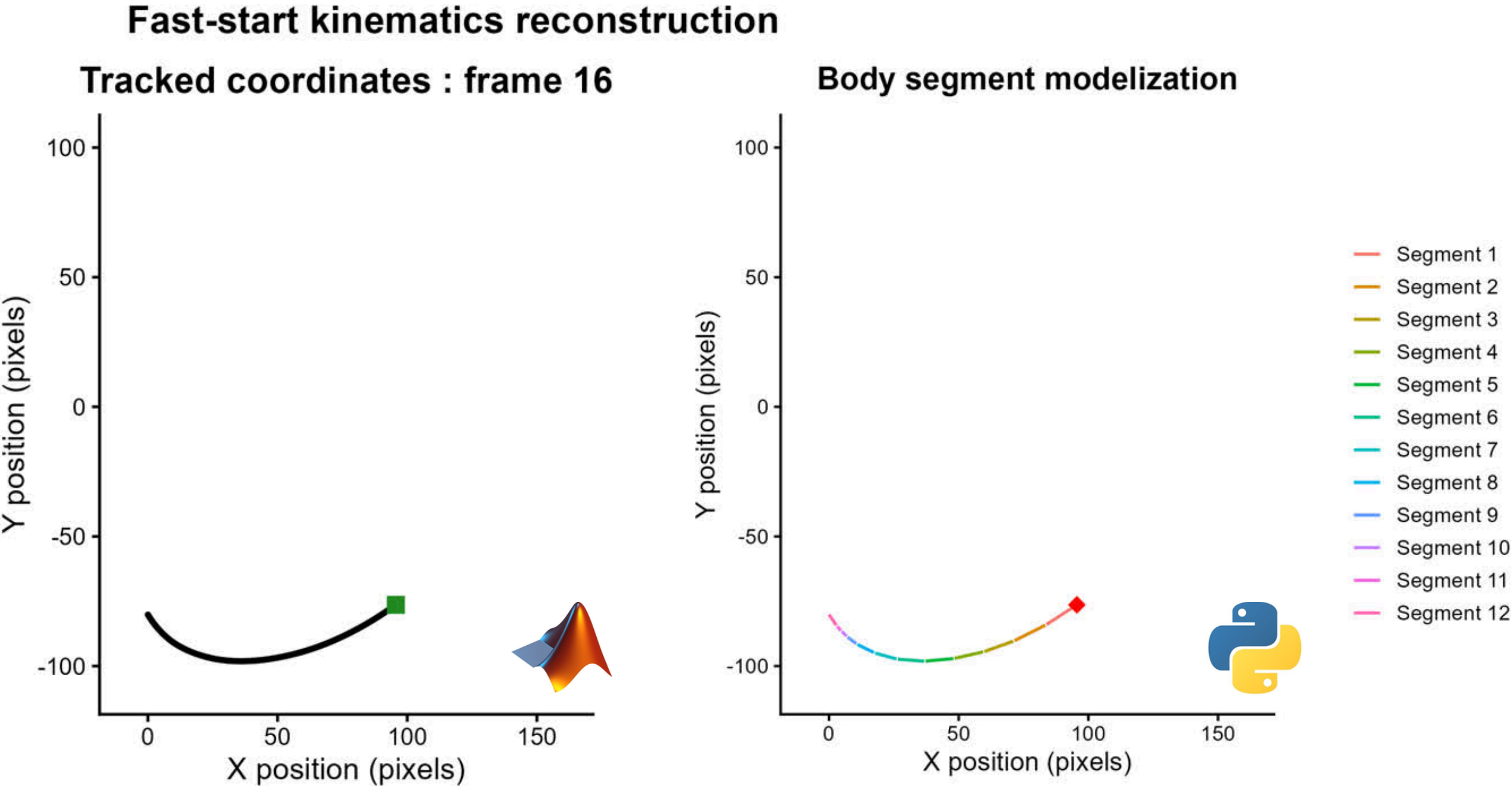
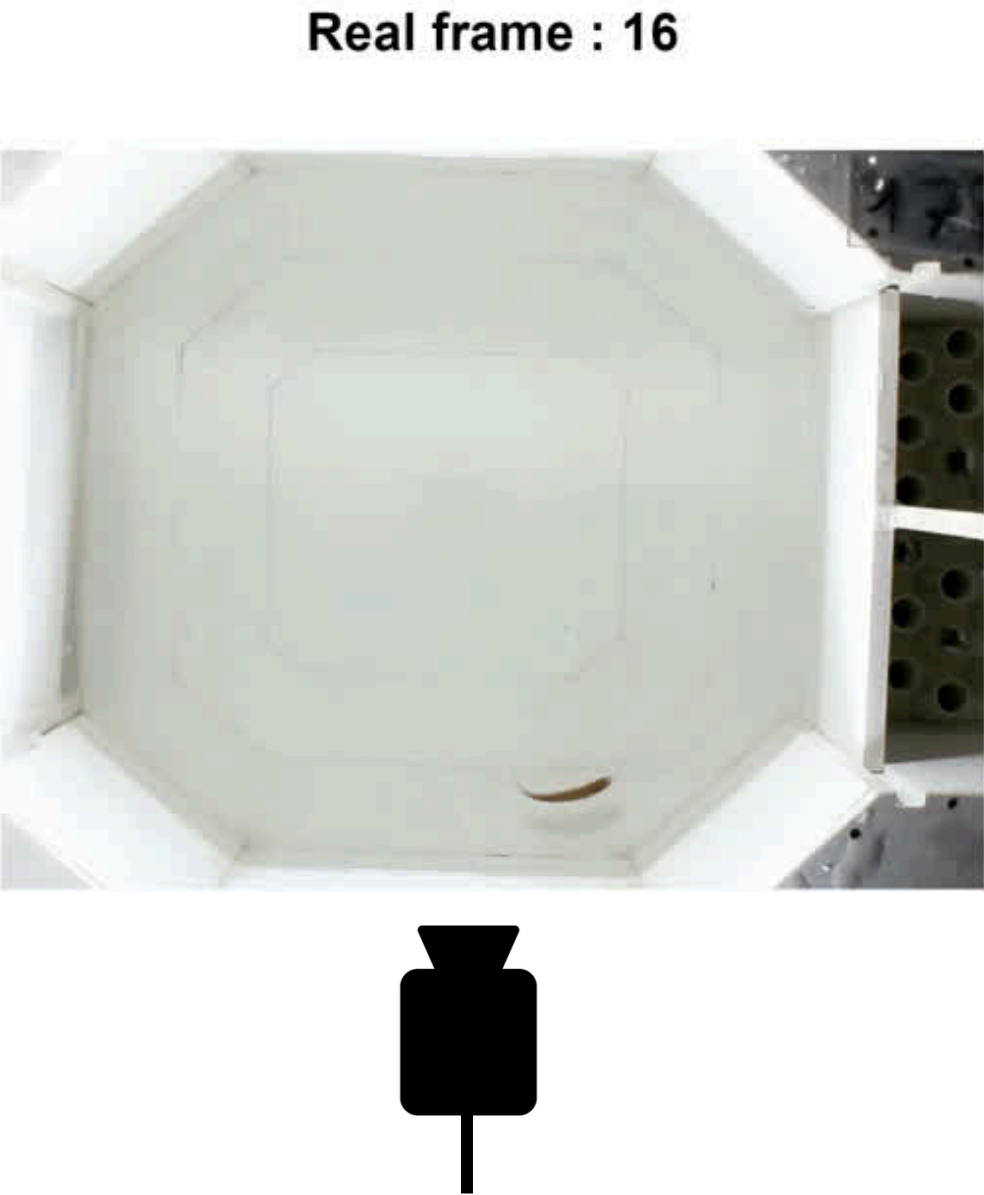


Figure 5: Illustration of the digitalization process used for kinematics analysis



Analysis:

Exploratory analysis - Random Forest
for unsteady behaviour analysis

- **No** strong **discriminative variable** to explain treatment.
- The variables with the most classifying potential were the **latency** to transition **zone**, the **difference** of head **movement**, and the **time spent** in center **zone**.

Modelling - General Additive Mixed Model (GAMM)
for time related metrics during fast start event



Statistical modeling - GA(M)M (General Additive Mixed Model).

Table 1: Comparison of the fixed effect selected for each model

Fixed effect	Metric measured during fast-start event				
	Distance cumulated	Acceleration	Segmentation	Angular ratio snout/tail	Angular ratio anterior/posterior
Treatment					
Genetic origin					
Size					

Significant effect

Undetected in the model

- No strong effect of **treatment** detected.
- Different temporal dynamics depending on the **genetic origin**.



Statistical modeling - GA(M)M (General Additive Mixed Model)

e.g. cumulated distance achieved

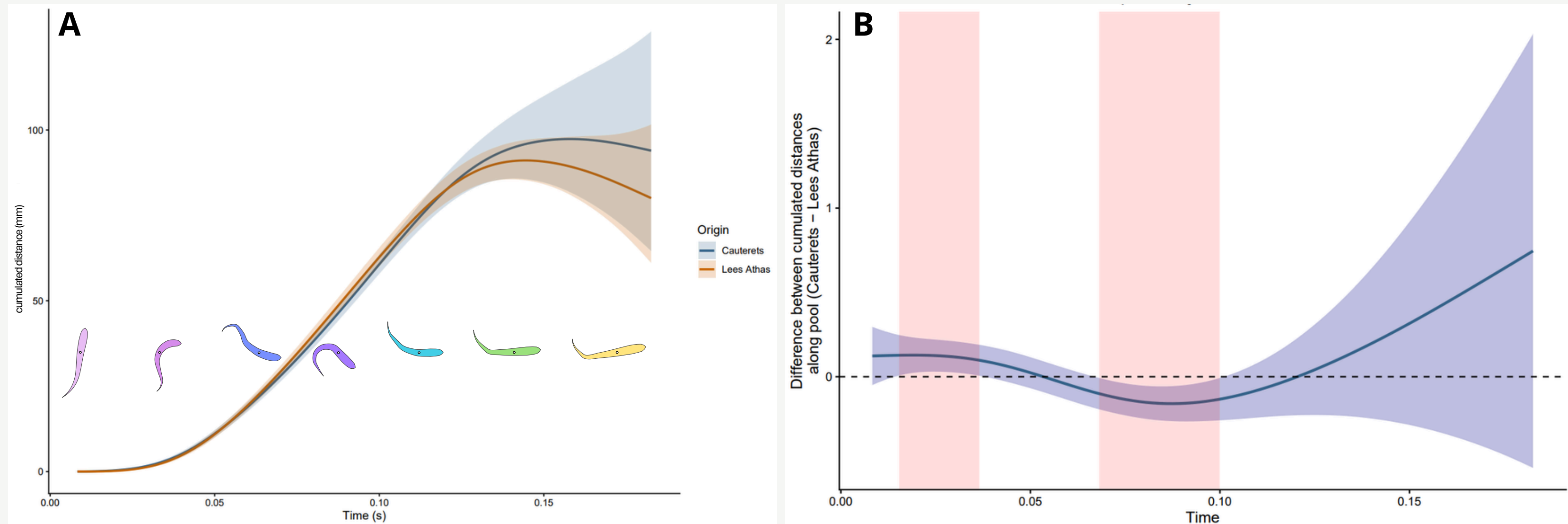


Figure 6: **A)** Predictions of non-linear relations between the evolution of the cumulated distance achieved during fast start across time ($\Delta\text{frame} = 0.0083\text{s}$) for the two genetic origins. **B)** Estimated difference between the temporal trajectories of the cumulated distance achieved along the pool (Cauterets - Lees Athas). Significant temporal differences are highlighted in red, where the IC95% difference does not include 0. Silhouettes are showed to describe actuation of the fast-start.



Statistical modeling - GA(M)M (General Additive Mixed Model)

e.g. cumulated distance achieved

Cauterets individuals achieve **more distance** at the start [15;35]ms.

Differences during the **early stages** of fast start [15;35]ms and [70;100]ms.

→ No effect of treatment detected.

→ Even the smallest difference can impact the surviving probability.

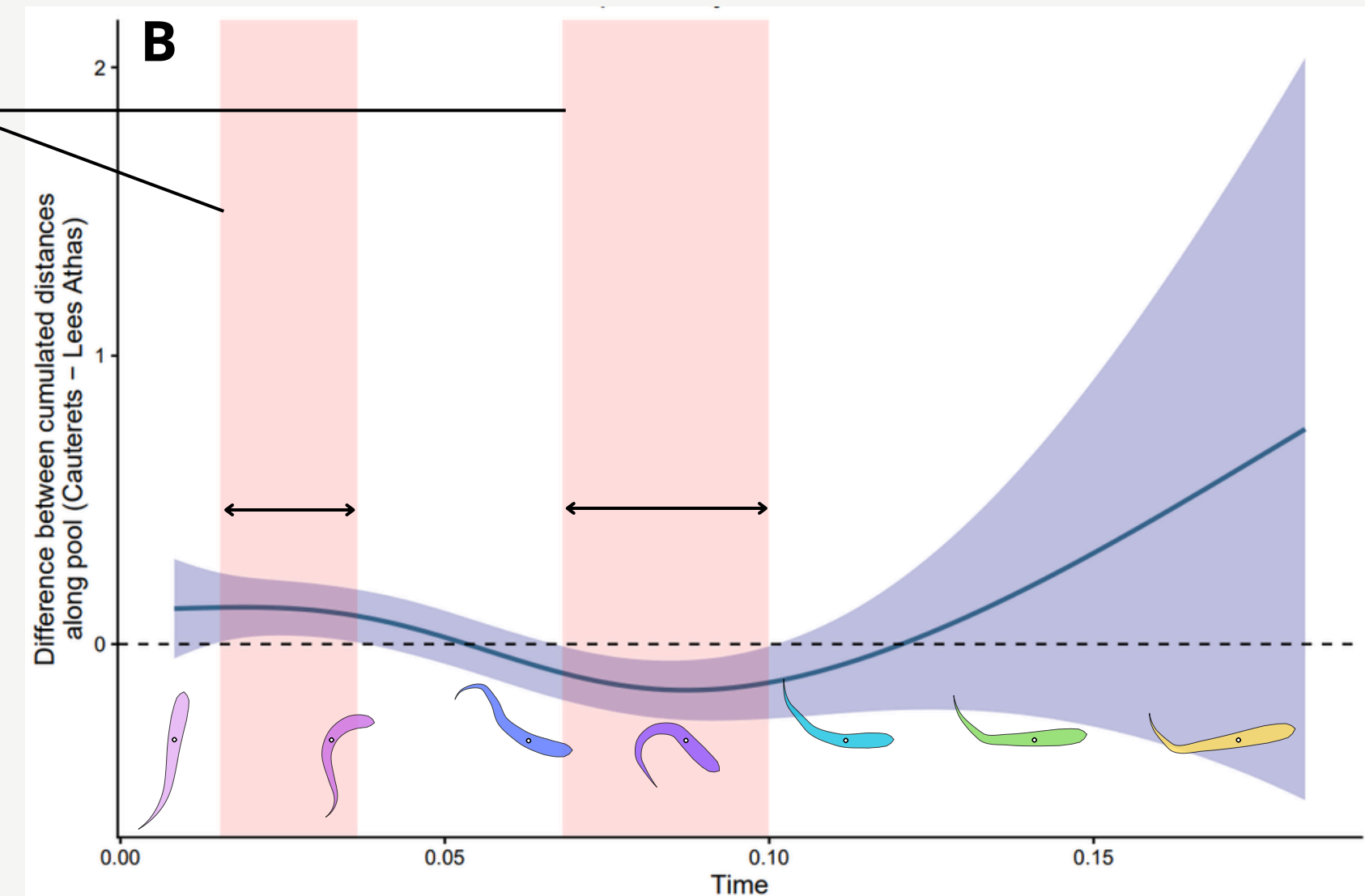
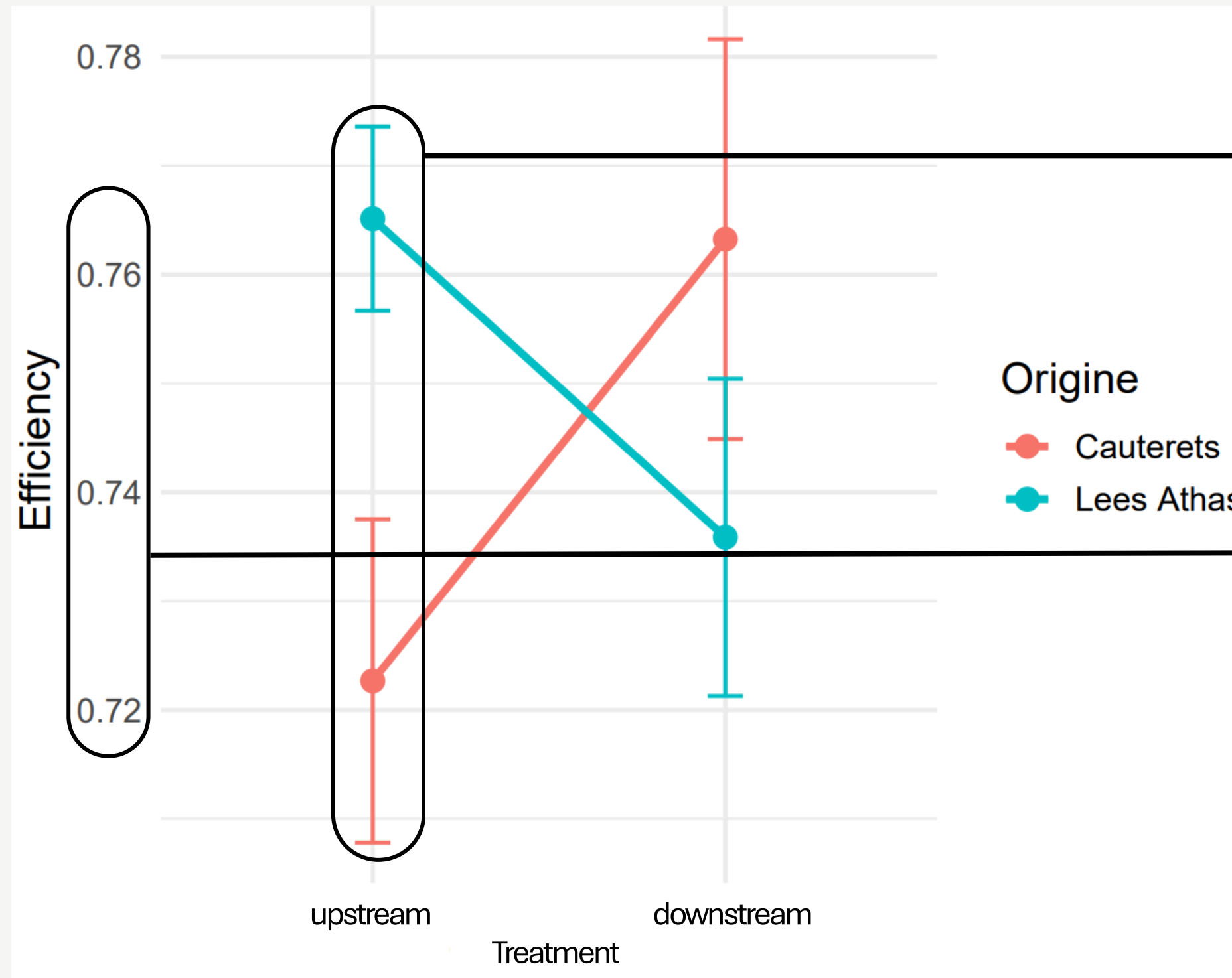


Figure 6: **A)** Predictions of non-linear relations between the evolution of the cumulated distance achieved during fast start across time ($\Delta\text{frame} = 0.0083\text{s}$) for the two genetic origins. **B)** Estimated difference between the temporal trajectories of the cumulated distance achieved along the pool (Cauterets - Lees Athas). Significant temporal differences are highlighted in red, where the IC95% difference does not include 0. Silhouettes are showed to describe actuation of the fast-start.

Statistical analysis - GLS (General Least Squares)

e.g. Efficiency

Differences ? Yes but how much ?



Upstream conditions : **varying flow**

Lees Athas : **Reared individuals**

→ Better efficiency

0.04 difference in terms of efficiency (~**0.4 cm**)

Do faster starts increase the probability of evading predators?

J. A. WALKER,*† C. K. GHALAMBOR,‡ O. L. GRISET,* D. McKENNEY* and D. N. REZNICK§

"On average, a one standard deviation increase in fast-start performance increases the odds of surviving a predation strike 2.3-fold"

Figure 7: Representation of the estimated values of efficiency and IC95% obtained from the GLS model comparing the efficiency during fast start.

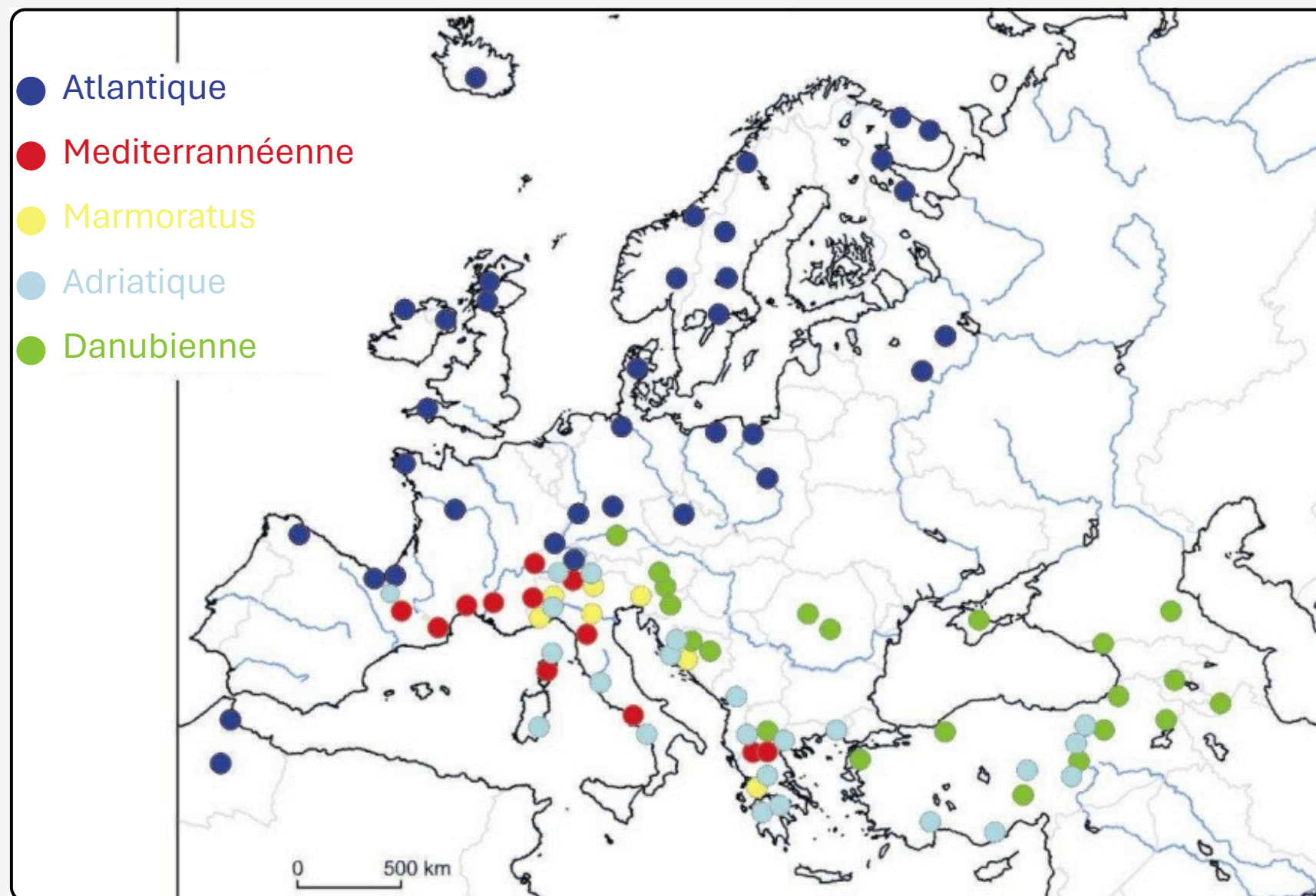
Importance of the genetic lineage:

Cauterets:

Mediterranean **wild** pool
collected by FDAAPPMA 65

Lees Athas:

Atlantic **rear** pool collected
from pisciculture



- Wild individuals show higher flexibility and burst performance.
- Possible domestical syndrome (30 years old pisciculture).
- Importance for repopulation.

Figure 8: European spatial distribution of the five genetic lineages of *S. trutta*. Adapted from Bernatchez, 2001.

Effect of the fluctuant flow regime:

Intensity / frequency of the flow variation ?

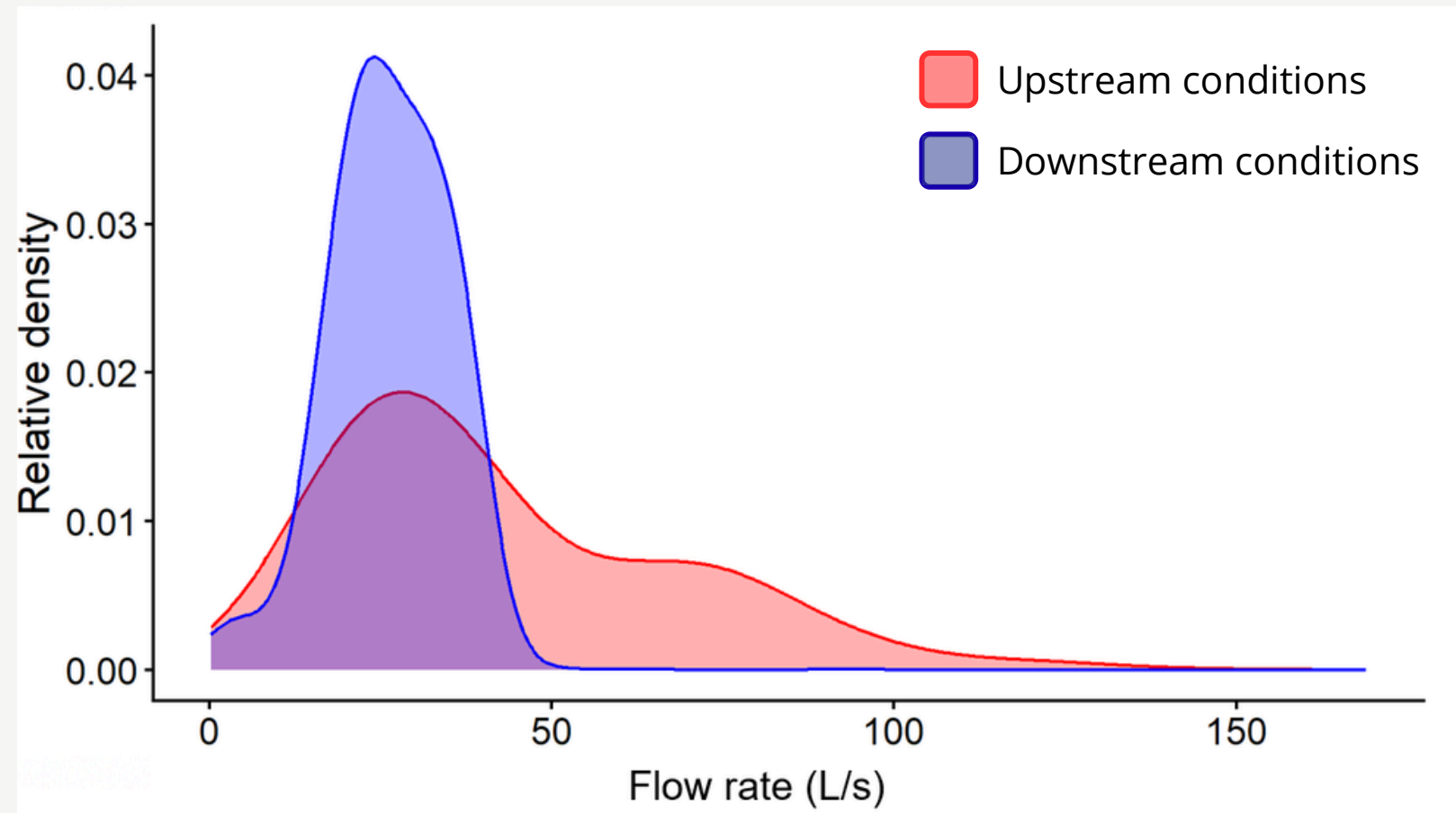
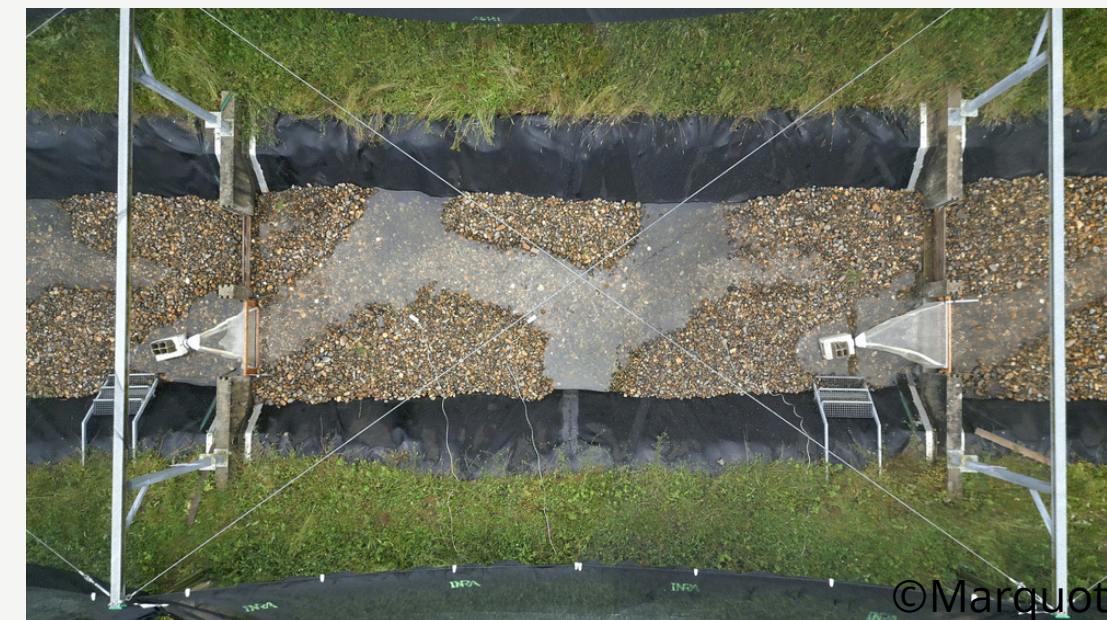


Figure 9: Density of the flow regime distribution for each treatment conditions.

→ Small compared to 'once-in-a-century flood'.

→ No test of droughts.

Experimental design limitations ?



→ Semi-natural complexity, refuges, microniches.

→ High robustness of individuals.

Effect of the fluctuant flow regime:

Morphology variability

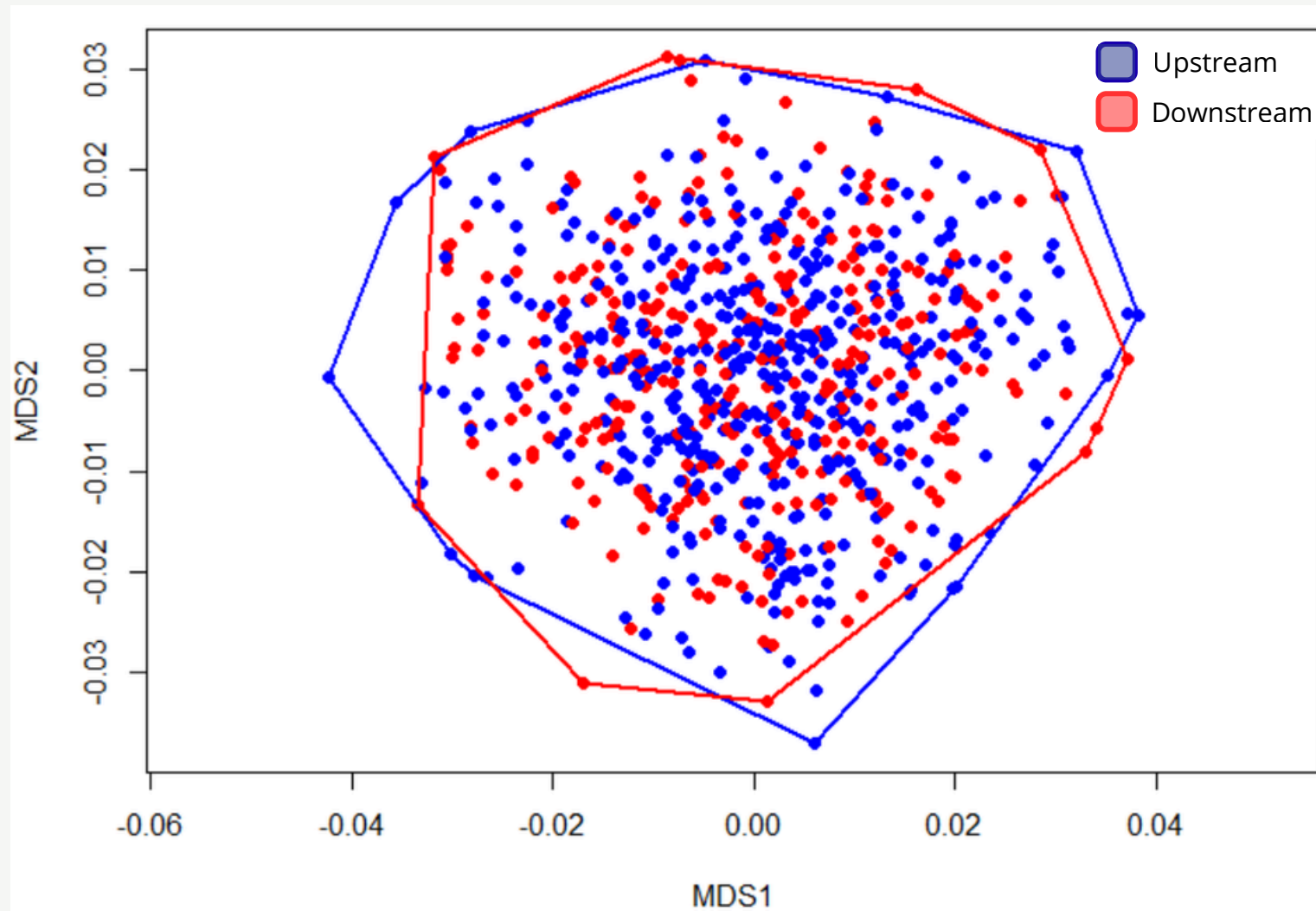
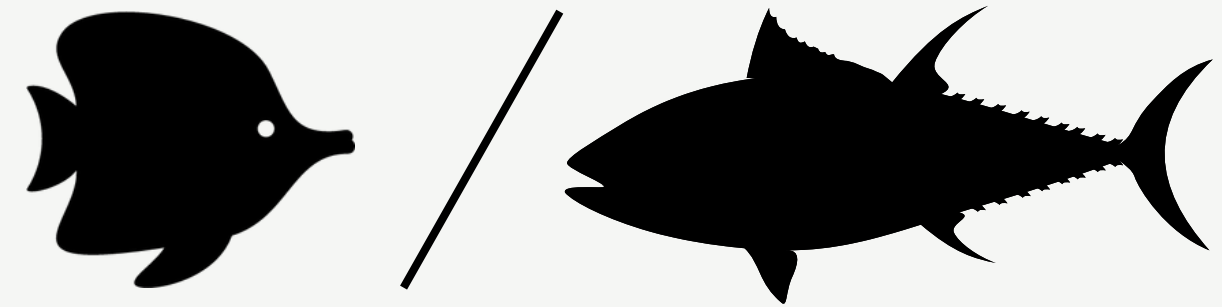


Figure 10: Illustration of the morphospace depending on the treatment conditions (stress=0.178). Ongoing PhD project.

→ No effect of varying conditions.

Trade-off comparison



→ Supplementary analysis showed no significant link between the steady/unsteady analysis.

→ Low number of observations.

Impact of **hydrological events** (floods) on swimming **performances** of *Salmonidae*

- **Development of fishes** in environments **contrasting in flow regimes** should **not influence** their **performances** (with our parameters/conditions) and shouldn't constraint nor **locomotion** capacities, nor **morphotypes**.
- It could be that **genetic importance** in development is much **stronger** than our environmental conditions resulting in the **absence** of significant differences linked to the **treatment**, suggesting high issues for **repopulation actions**.
- Only **early stages** of life have been investigated, but possible variation could occur and be detected **throughout the life** in terms of **behaviour** or **performance**, or even in descendants traits.
- Observed **developmental pattern** seems to show **robustness** in **varying environment** with refuges, and suggest **persistence of phenotype** in stressful conditions.

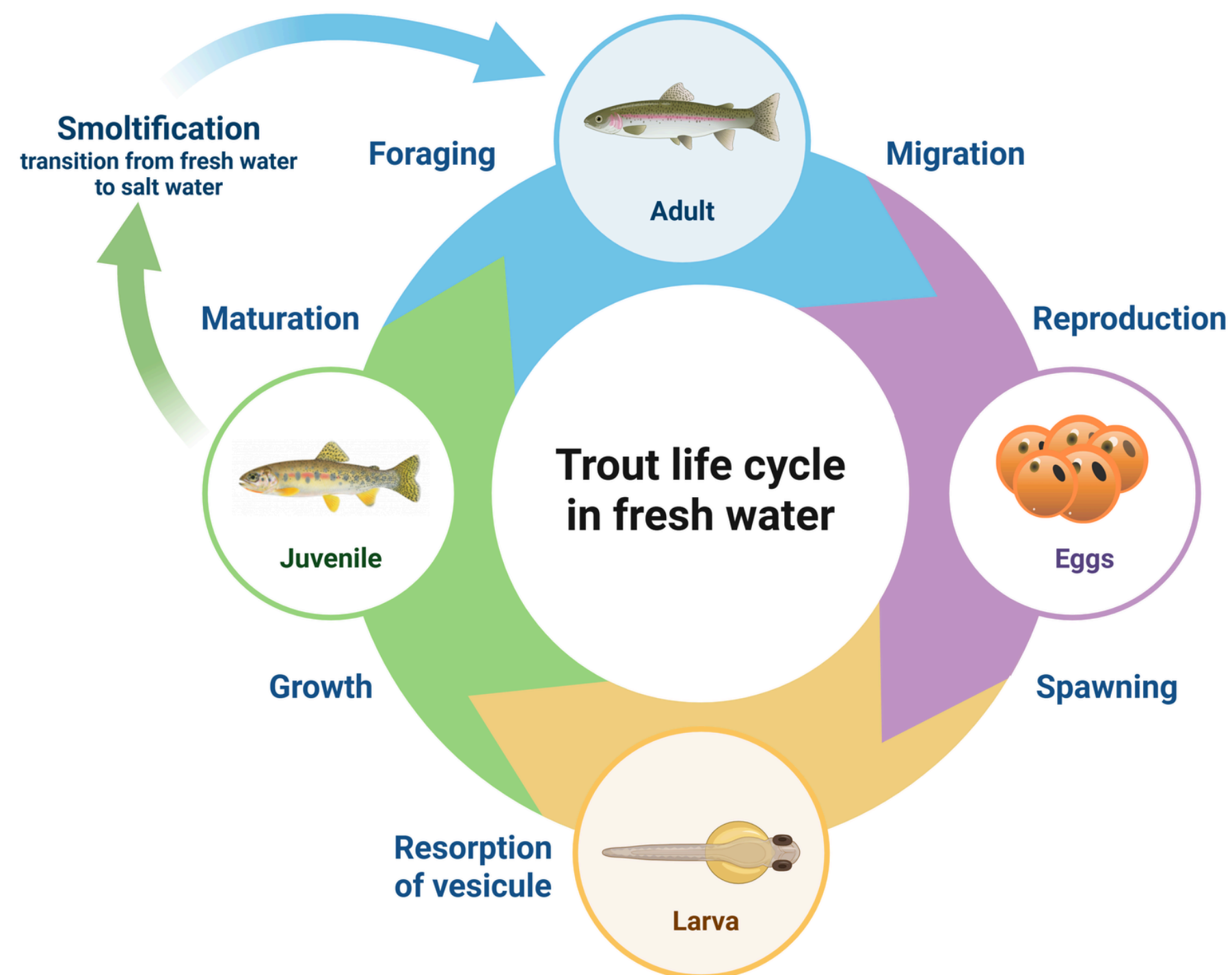
Thank you for listening



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Biological model:



Exploratory analysis

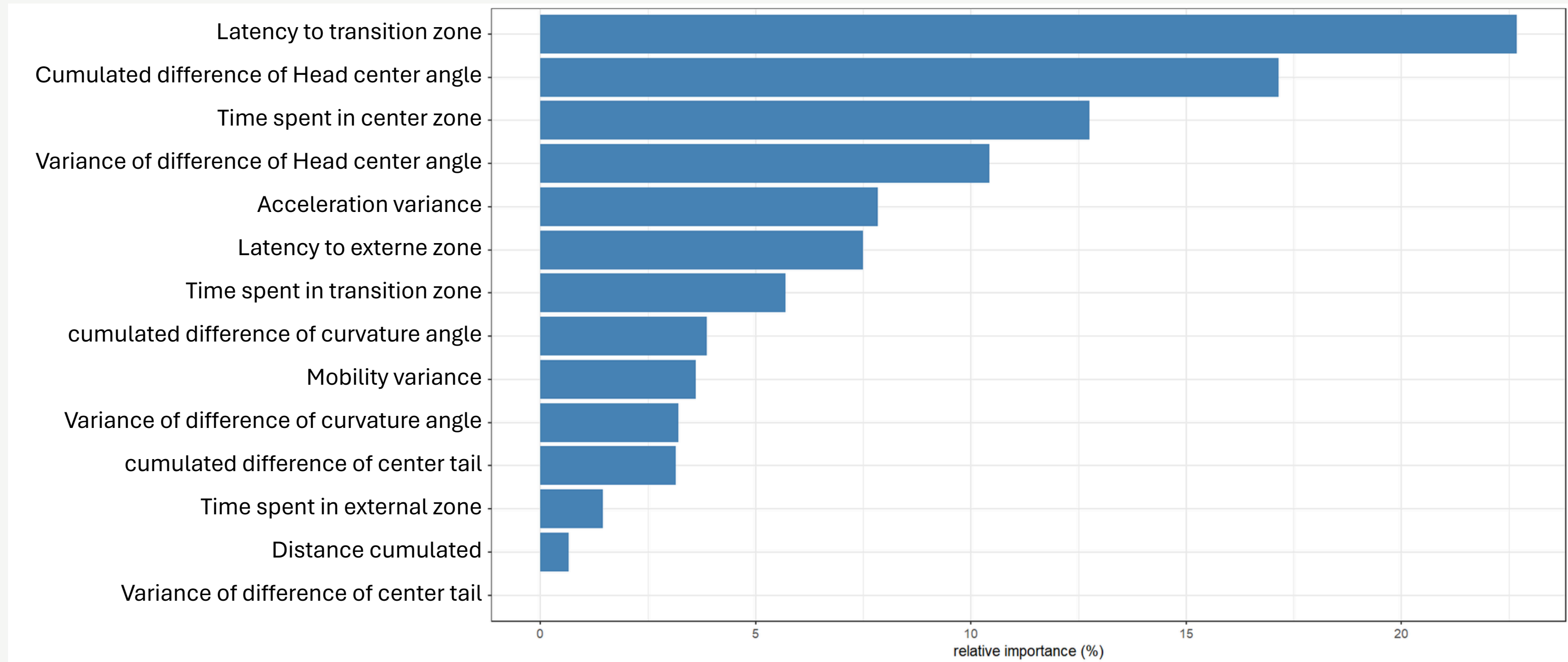


Figure 6: Relative importance of each variable in in the classification of 'treatment' in Random Forest models (ntree = 5000). Relative importance is calculated from MDA values.

- No strong discriminative variable to explain treatment
- No significative differences between groups for the first variable

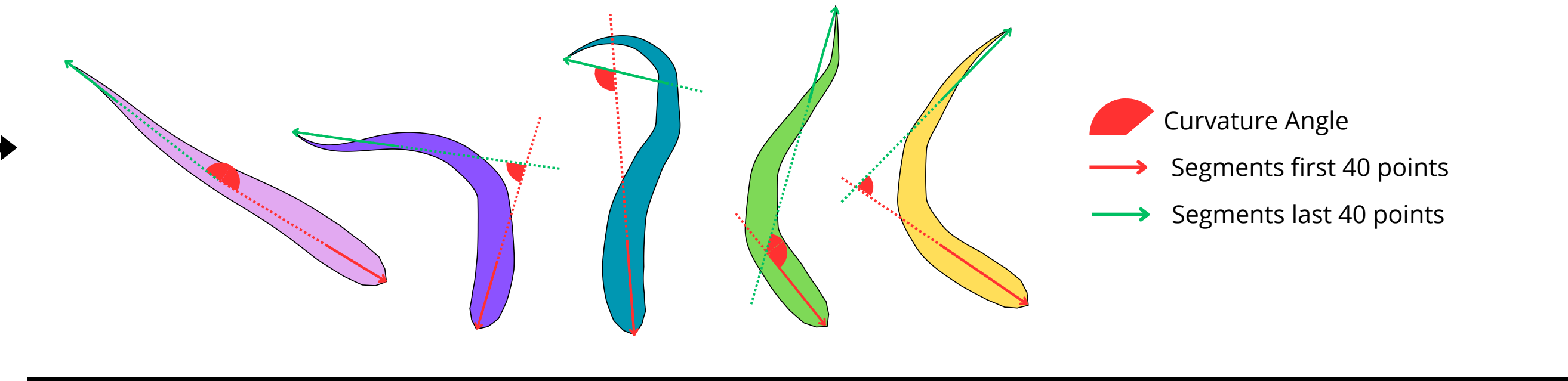
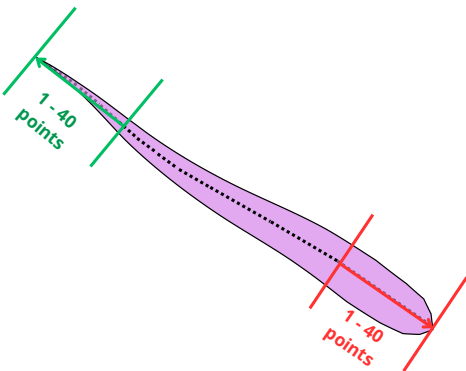
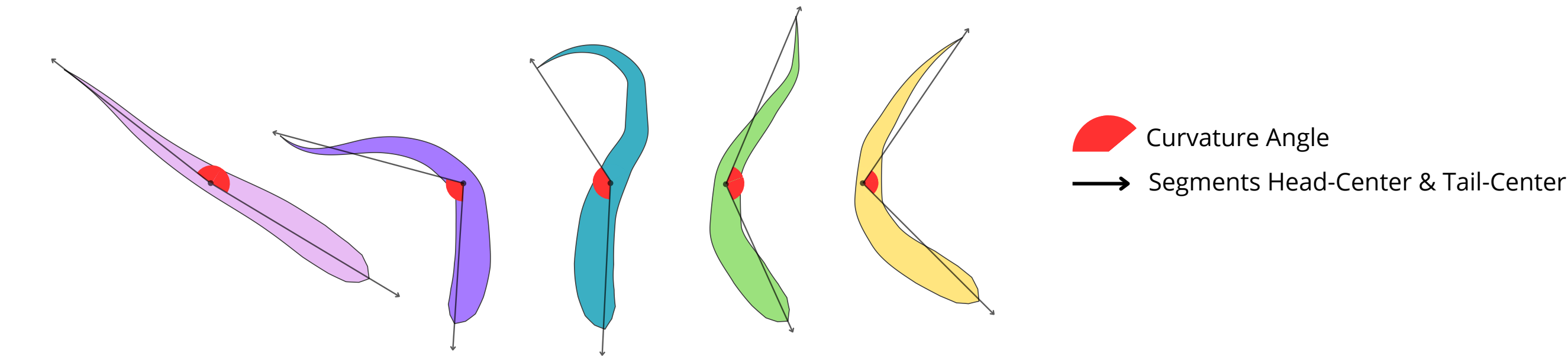
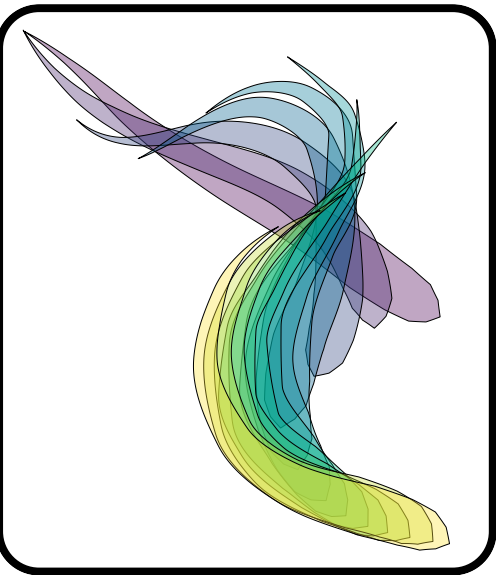
Table of GA(M)M selection for the cumulated distance achieved

model_type	formula	(e)df	AIC	deltaAIC	weight
GAM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + s(\text{time}) + s(\text{fs}, \text{bs} = "re")$	192,0631354	6419,064422	0	0,159565485
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = "re") + s(\text{fs}, \text{bs} = "re")$	191,8368416	6419,504578	0,440156558	0,128044278
GAM	$Y \sim s(\text{time}, \text{pool}, \text{bs} = "fs") + s(\text{fs}, \text{bs} = "re")$	192,6318581	6419,862359	0,797937344	0,107070311
GAM	$Y \sim \text{treatment} + \text{pool} + \text{bief} + \text{origin_treatment} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = "re") + s(\text{fs}, \text{bs} = "re")$	192,2617842	6419,909574	0,845152596	0,104572238
GAM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + s(\text{fs}, \text{bs} = "re")$	192,8692376	6420,614425	1,550003272	0,073512302
GAM	$Y \sim \text{pool} + s(\text{time}, \text{by} = \text{pool}) + s(\text{fs}, \text{bs} = "re")$	192,9872092	6420,697014	1,632592598	0,070538461
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{fs}, \text{bs} = "re")$	192,9549277	6420,731037	1,666615696	0,069348641
GAM	$Y \sim \text{treatment} + \text{pool} + s(\text{time}, \text{by} = \text{pool}) + s(\text{fs}, \text{bs} = "re")$	193,0730758	6420,787704	1,723282705	0,067411326
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + s(\text{time}, \text{by} = \text{treatment}) + s(\text{poisson}, \text{bs} = "re") + s(\text{fs}, \text{bs} = "re")$	192,1487479	6420,880986	1,816564204	0,064339407
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = "re") + s(\text{fs}, \text{bs} = "re")$	192,6600833	6420,971494	1,907072231	0,061492689
GAM	$Y \sim \text{treatment} + \text{pool} + \text{temperature_ellerby} + \text{origin_treatment} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = "re") + s(\text{fs}, \text{bs} = "re")$	194,0456779	6421,798312	2,733890402	0,040670752
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}) + s(\text{poisson}, \text{bs} = "re") + s(\text{fs}, \text{bs} = "re")$	187,0778649	6423,39025	4,325828184	0,018348362
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}) + s(\text{poisson}, \text{bs} = "re") + s(\text{poisson_fs}, \text{bs} = "re")$	187,0778649	6423,39025	4,325828184	0,018348362
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + s(\text{time}) + s(\text{poisson}, \text{bs} = "re") + s(\text{fs}, \text{bs} = "re")$	187,0777162	6423,577738	4,513316232	0,016706474
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}) + \text{poisson} + \text{fs}$	200,2262394	6436,16266	17,09823788	3,09103E-05
GAM	$Y \sim s(\text{time}) + s(\text{fs}, \text{bs} = "re")$	190,0667063	6578,580453	159,5160313	3,66837E-36
GAMM	$Y \sim s(\text{time}) + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	10	6804,571181	385,5067594	3,09892E-85
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) \mid \text{random} = \text{fs}$	7	6810,171014	391,1065923	1,88461E-86
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{pool} \mid \text{random} = \text{fs}$	8	6813,878664	394,8142421	2,95199E-87
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	12	6815,688152	396,6237299	1,19451E-87
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{treatment} + \text{pool} \mid \text{random} = \text{fs}$	9	6817,046094	397,9816722	6,05783E-88
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{fs} + \text{poisson}$	11	6817,893706	398,8292839	3,96516E-88
GAMM	$Y \sim s(\text{time}, \text{by} = \text{treatment}) + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	12	6818,123309	399,0588872	3,53511E-88
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{fs}$	10	6823,544725	404,4803035	2,35048E-89
GAMM	$Y \sim s(\text{time}) \mid \text{random} = \text{fs}$	5	6959,582138	540,5177164	6,7765E-119
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = "re")$	80,33673466	7449,552712	1030,48829	2,7243E-225
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{poisson}$	10	7622,916987	1203,852565	6,1614E-263

Table of GA(M)M selection for all models

REML					Effects selected in the GAM model							Random Effect		
AIC weight scale maximum weight minimum weight				0.000	0.001	Time (Origin) +	Treatment +	Origin +	Size +	Temperature +	Bief +	Interaction (Origin/Treatment) +	RE (fast-start)	RE (fish)
	0.041	0.000	0.021 ^(s)	0.000	0.001	Time (Origin) +	Treatment +	Origin +	Size +	Temperature +		Interaction (Origin/Treatment) +	RE (fast-start)	RE (fish)
	0.105	0.000	0.147 ^(s)	0.027	0.048	Time (Origin) +	Treatment +	Origin +	Size +		Bief +	Interaction (Origin/Treatment) +	RE (fast-start)	RE (fish)
				0.000	0.000	Time (Size) +	Treatment +	Origin +	Size +			Interaction (Origin/Treatment) +	RE (fast-start)	RE (fish)
				0.020	0.000	Time (Origin) +	Treatment +	Origin +	Size +		Bief +		RE (fast-start)	RE (fish)
	0.064	0.000	0.000 ^(s)	0.000	0.000	Time (Treatment) +	Treatment +	Origin +	Size +			Interaction (Origin/Treatment) +	RE (fast-start)	RE (fish)
	0.128	0.000	0.152 ^(s)	0.130	0.000	Time (Origin) +	Treatment +	Origin +	Size +			Interaction (Origin/Treatment) +	RE (fast-start)	RE (fish)
				0.047	0.000	Time (Origin) +		Origin +	Size +		Bief +		RE (fast-start)	RE (fish)
	0.017	0.000	0.000 ^(s)	0.000	0.000	Time +	Treatment +	Origin +	Size +			Interaction (Origin/Treatment) +	RE (fast-start)	RE (fish)
			0.005 ^(s)			Time (Origin) +		Origin +	Size +			+ Time	RE (fast-start)	
	0.069		0.142 ^(s)			Time (Origin) +	Treatment +	Origin +	Size +				RE (fast-start)	
				0.056	0.143	Time (Origin) +		Origin +			Bief +		RE (fast-start)	
	0.160			0.364	0.065	Time (Origin) +						+ Time	RE (fast-start)	
	0.067					Time (Origin) +	Treatment +	Origin +					RE (fast-start)	
	0.018	0.000	0.000 ^(s)			Time +	Treatment +	Origin +	Size +					RE (fish)
			0.109 ^(s)			Time (Origin *) +	Treatment +	Origin +					RE (fast-start)	
			0.147 ^(s)			Time (Origin) +		Origin +	Size +				RE (fast-start)	
	0.107			0.067	0.300	Time (Origin *) +		Origin +					RE (fast-start)	
		0.000				Time +	Treatment +	Origin +					RE (fast-start)	
		0.002				Time (Treatment) +	Treatment +						RE (fast-start)	
	0.074	0.000	0.086 ^(s)	0.215	0.442	Time (Origin) +							RE (fast-start)	
		0.000				Time +		Origin +						
		0.998				Time +								
Distance cumulated ⁽¹⁾					(*) shared wiggliness									
Acceleration ⁽²⁾					(s) + segment)									
Segmentation ⁽³⁾														
Ratio snout/tail ⁽⁴⁾														
Ratio ant/post ⁽⁵⁾														

Illustration of the evolution of the angular metrics per frame during a fast-start event



Time (frame)

frame 01 - 00ms

frame 03 - 16ms

frame 05 - 32ms

frame 07 - 48ms

frame 09 - 64ms

